Near-Field Channel Estimation for XL-RIS Assisted Multi-User XL-MIMO Systems: Hybrid Beamforming Architectures

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***Abstract*— Reconfigurable intelligent surface (RIS) is an emerging technique for robust millimeter-wave (mmWave) multiple-input multiple-output (MIMO) systems. In this paper, we study the channel estimation problem for extremely large-scale RIS (XL-RIS) assisted multi-user XL-MIMO systems with hybrid beamforming structures. In this system, we propose an *unified* channel estimation method that yields a notable estimation accuracy in the near-field BS-RIS and near-field RIS-User channels (in short, near-near field channels), far-near field channels, and far-far field channels. Our key idea is that the effective channels to be estimated can be each factorized as the product of low-rank matrices (i.e., the product of a common matrix and a user-specific coefficient matrix). The common matrix whose columns are the basis of the column space of the BS-RIS channel is efficiently estimated via a *collaborative* low-rank approximation (CLRA). Leveraging the hybrid beamforming structures, we develop an efficient iterative algorithm that jointly optimizes the user-specific coefficient matrices. Via experiments and complexity analysis, we verify the effectiveness of the proposed channel estimation method (named CLRA-JO) for the three categories of wireless channels.**

***Index Terms*— Reconfigurable intelligent surface (RIS), XL-MIMO, channel estimation, hybrid beamforming, low-rank approximation.**

I. Introduction

RECONFIGURABLE intelligent surface (RIS) is a promising technique for robust millimeter-wave (mmWave) and terahertz (THz) multiple-input multiple-output (MIMO) systems [1], [2], [3]. A RIS consists of a uniform array with a large number of reflective elements, each of which can control the phase and reflection angle of the incident signal so that the received power of the intended signal is enhanced [4]. The potential advantages of the RIS open up new research opportunities such as reflect beamforming design [5], [6], [7], [8] and RIS-aided localization and sensing [9], [10], [11]. The accuracy of a channel estimation plays a crucial role in implementing various RIS-aided applications [11].

There have been numerous works on the design of an efficient channel estimation method in RIS-aided mmWave MIMO systems. Herein, the channel responses between the base station (BS) and the RIS (in short, the BS-RIS channel) and between the RIS and the users (in short, the RIS-User channels) are assumed as *far-field* channels. Accordingly, both BS and RIS will experience *planar-wavefront*. The ultimate goal of the existing works is to estimate the so-called effective channel accurately while having an affordable training overhead. To accomplish this, the key idea is to exploit the *sparsity* of mmWave channels, which comes from the fact that there exists a smaller number of signal paths in the BS-RIS channel. The de-facto channel estimation method to harness the sparsity is based on compressed sensing (CS). Specifically, the angle-of-departures (AoDs) and the angle-of-arrivals (AoAs) of sparse signal paths are estimated via CS thereby being able to require less training overhead. In [12], the popular orthogonal matching pursuit (OMP) was adopted as a low-complexity CS algorithm. An enhanced CS-based channel estimation method was presented in [13] by exploiting the subspace estimation and the so-called scaling property of the RIS-User channels. However, the CS-based methods suffer from the inevitable *grid-mismatch* problem [14] since a dictionary matrix is constructed by quantizing steering vectors at a specific resolution. Having an affordable computational complexity (i.e., the quantization levels), they result in a severe error-floor problem due to the quantization errors.

Approximate message passing (AMP)-based channel estimation methods were developed in [15] and [16], by employing the parallel factor decomposition to unfold the effective channels. In [15], an alternating least squares (ALS) and a vector AMP (VAMP) were presented as efficient iterative

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algorithms. Also, in [16], a unitary AMP (UAMP)-based method was presented, which is more computationally efficient than the ALS- and VAMP-based methods. Noticeably, the AMP-based methods are only applicable in larger systems such that the number of users (denoted by K) is not less than the number of reflective elements (denoted by N) in the RIS. In the example of $N = 128$, the AMP-based methods can be used at least when $K \geq 128$. This restricts the applicability of the AMP-based methods in various RIS-aided applications.

In future communication systems (e.g., 6G), to further enhance the beamforming gain, the number of reflective elements in the RIS tends to be extremely large (XL) [17], [18]. This will lead to a larger Rayleigh distance (RD), which is the boundary between the near- and far-field channels. For instance, when the RIS is equipped with 256 half-wavelength spacing elements at operating $100\mathrm{GHz}$, the RD is almost $100\mathrm{m}$ [19]. In the 6G systems, therefore, the near-field effect becomes non-negligible and consequently the RIS will experience *spherical-wavefront* rather than planar-wavefront (see Fig. 2(b)). We note that the aforementioned channel estimation methods based on the planar-wavefront heavily rely on the channel sparsity in the angular domain. Due to the near-field effect in the spherical-wavefront, however, the channel sparsity in the angular domain is no longer applicable. It is necessary to develop an efficient channel estimation method for near-field XL-RIS assisted MIMO systems. In [17] and [20], the channel estimation problem was formulated as a CS problem by means of the polar-domain sparsity. Based on this framework, a single measurement vector (SMV)-based OMP was presented as an efficient channel estimation method. As expected, this approach suffers from the grid-mismatch problem.

In the above researches, it is assumed that a BS is equipped with fully-digital structures. However, when the BS is equipped with a massive or extremely large number of antennas, the use of a large number of radio frequency (RF) chains (i.e., one for each antenna component) significantly increases the cost and energy consumption [21], [22]. One promising solution to these problems lies in the concept of *hybrid* beamforming structures [22], which combines an analog beamforming in the RF domain and a digital beamforming in the baseband connected to the RF chains. The XL-RIS assisted XL-MIMO systems with hybrid beamforming structures are considered as a promising energy-efficiency technique [23], [24], [25]. Recently in [26], [27], [28], and [29], the channel estimation method for RIS-aided mmWave MIMO systems with hybrid beamforming structures was investigated, in which the BS-RIS and RIS-User channels are assumed as *far-field* channels. In [26] and [27], the grid-mismatch problem was alleviated via an atomic norm minimization (ANM). However, the ANM-based methods are impractical due to their unaffordable computational complexities. Recently in [28] and [29], the channel estimation problem was formulated as a low-rank matrix completion (LRMC) and then solved via a fast alternating least squares (FALS) [30]. Also, it was shown that the LRMC-based method can outperform the the CS-based methods by avoiding the grid-mismatch problem [28], [29]. Since the matrix completion is performed from *noisy* observations, the LRMC-based method suffers from the *noisy-sample* problem. Specifically, the sampled noisy elements cannot be updated via the matrix completion method proposed in [28] and [29]. Consequently, an estimation accuracy is saturated even if the pilot overhead (i.e., the number of sampled elements) increases. In very low signal-to-noise ratio (SNR) regimes, the LRMC-based method cannot yield an attractive estimation accuracy. In [31] and [32], the XL-RIS assisted MU-MIMO systems with hybrid beamforming structures, which is the most related to our system model, was investigated. Herein, the BS-RIS and the RIS-User channels are assumed as *far- and near-field* channels, respectively. Due to the use of hybrid beamforming structures, the BS can observe a smaller number of training measurements, which naturally increases the training overhead to achieve a similar estimation accuracy with the fully-digital counterpart. In [31], a two-phase channel estimation protocol was proposed, which can improve the efficiency of training overhead. Based on this framework, a 3-dimensional multiple measurement vector (MMV) look-ahead OMP (3D-MLAOMP) was developed as an efficient channel estimation method. This method still suffers from the grid-mismatch problem. Moreover, it requires some impractical assumptions such as the number of spatial paths and the angle of line-of-sight (LoS) path being given as a priori. In [32], a fast-sparse-Bayesian-learning (FSBL)-based channel estimation method was proposed by focusing on the limited visual region of the channel between the RIS and each user in XL-RIS systems. This method is impractical due to the expensive computational complexity since a super-resolution (SR) algorithm is used to estimate the common AoD at the BS and its complexity is too expensive. Therefore, it is an open problem to construct an efficient channel estimation method for XL-RIS assisted XL-MIMO systems.

Motivated by the above, we in this paper study the channel estimation problem for XL-RIS assisted multi-user XL-MIMO systems with hybrid beamforming structures. According to the distances between the BS and the RIS, and between the RIS and the users, the corresponding wireless channels are categorized such as

1) Far-field BS-RIS and far-field RIS-User channels (in short, *far-far* field channel);
2) Far-field BS-RIS and near-field RIS-User channels (in short, *far-near* field channel);
3) Near-field BS-RIS and near-field RIS-User channels (in short, *near-near* field channel).

Beyond the existing works in [31] and [32], we for the first time investigate the channel estimation problem for near-near field channels. Furthermore, as an extension of the existing works, the proposed channel estimation method can be performed when each user is equipped with multiple antennas. The major contributions of this paper are summarized as follows:

- We propose an *unified* channel estimation method, which can be applied to the aforementioned three category channels without any modification. Whereas, in the state-of-the-art (SOTA) methods [13], [31], dictionary matrices should be constructed according to the channel categories. To further reduce the training overhead, we present a

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two-phase communication protocol suitable for the proposed channel estimation method.

- The proposed method is constructed by means of the low-rank factorization of effective channels (denoted by $\{\mathbf{H}_{[\text{eff},k]} : k \in [K]\}$): they can be decomposed into the product of low-rank matrices, i.e., $\mathbf{H}_{[\text{eff},k]} = \mathbf{S}_{\text{col}}\mathbf{T}_k$, where $\mathbf{S}_{\text{col}}$ contains the basis of the *common* column space and $\mathbf{T}_k$ is a user-specific coefficient matrix.
- Harnessing the low-rank factorization, the proposed channel estimation method consists of the two parts. In the first part, the dimension of the column space (i.e., the size of $\mathbf{S}_{\text{col}}$) is estimated using the minimum description length (MDL) criterion and then $\mathbf{S}_{\text{col}}$ is estimated via a collaborative low-rank approximation (CLRA). We theoretically prove that the accuracy of the estimated $\mathbf{S}_{\text{col}}$ is inversely proportional to the product of the pilot overhead and the number of users. Thus, the proposed method can achieve a multi-user gain: given a desired accuracy, the pilot overhead can be reduced as the number of users grows.
- In the second part, we present an efficient iterative algorithm to *jointly* optimize the user-specific coefficient matrices $\{\mathbf{T}_k : k \in [K]\}$. The key idea is to employ the special property such as $\mathbf{H}_{[\text{eff},k]} = \mathbf{D}_k\mathbf{H}_{[\text{eff},1]}$, $\forall k \in [K]$, where $\mathbf{D}_k$ is a user-specific diagonal matrix. The convergence of our iterative algorithm is theoretically proved. Also, the superiority of our joint optimization compared with the standard *individual* least-square (LS) estimation is verified.
- The proposed channel estimation method is referred to as **C**ollaborative **L**ow-**R**ank **A**pproximation and **J**oint **O**ptimization (CLRA-JO). Via experiments and complexity analysis, we verify the effectiveness of the proposed CLRA-JO. In various channel environments, the proposed method can yield better estimation accuracy than the SOTA methods while having lower training overhead (e.g., about 80% reduction of the training overhead).

The remaining part of this paper is organized as follows. In Section II, we define the channel and signal models for XL-RIS assisted multi-user XL-MIMO systems with hybrid beamforming structures. Section III introduces the channel estimation protocol and the frame structures. In Section IV, we describe the proposed channel estimation method, named CLRA-JO. Section V provides simulation results and Section VI concludes the paper.

Notations. Let $[N_1 : N_2] \triangleq \{N_1, N_1+1, \ldots, N_2\}$ for any positive integers N_1 and N_2 with $N_1 < N_2$. For $N_1 = 1$, it is further simplified as $[N_2] \triangleq \{1, 2, \ldots, N_2\}$. We use $\mathbf{x}$ and $\mathbf{A}$ to denote a column vector and matrix, respectively. Also, $\mathbf{A}^\dagger$ denotes the Moore-Penrose inverse and $\otimes$ denotes the Kronecker product. Given a $M \times N$ matrix $\mathbf{A}$, let $\mathbf{A}(i,:)$ and $\mathbf{A}(:,j)$ be the i-th row and j-th column of $\mathbf{A}$, respectively, and $\|\mathbf{A}\|_F$ denotes the Frobenius norm of $\mathbf{A}$. Also, given the index subsets $\mathcal{I}_{\text{row}} \subseteq [M]$ and $\mathcal{I}_{\text{col}} \subseteq [N]$, let $\mathbf{A}(\mathcal{I}_{\text{row}},:)$ and $\mathbf{A}(:,\mathcal{I}_{\text{col}})$ be the submatrices of $\mathbf{A}$ by only taking the rows and columns whose indices are belong to $\mathcal{I}_{\text{row}}$ and $\mathcal{I}_{\text{col}}$, respectively. Given a vector $\mathbf{v}$, $\text{diag}(\mathbf{v})$ denotes a diagonal matrix whose ℓ-th diagonal element is equal to the ℓ-th element of $\mathbf{v}$ and $\|\mathbf{v}\|_2$ denotes the l_2-norm of $\mathbf{v}$. We let $\mathbf{I}$ and $\mathbf{0}$ denote the identity and all-zero matrices, respectively, where the sizes of these matrices are easily obtained from the context.

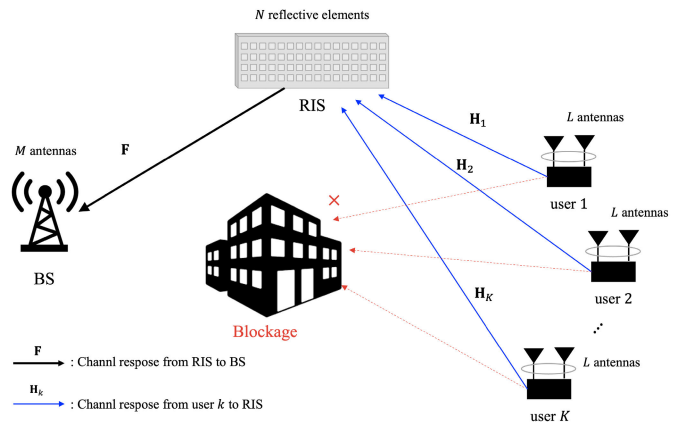

Fig. 1. A RIS-assisted MU-MIMO system consisting of the BS with M antennas, the RIS with N reflective elements, and the K users with L antennas.

II. System Model

We consider an uplink time-division-duplexing (TDD) multi-user multiple-input multiple-output (MU-MIMO) system with uniform planar arrays (UPAs). In this system, a reconfigurable intelligent surface (RIS) with N reflective elements assists the communication between the base station (BS) with M antennas and the K users each having L antennas (see Fig. 1). Taking the structure of UPA into account, throughout the paper, we let $N = N_h \times N_v$, $M = M_v \times M_h$, and $L = L_v \times L_h$. Noticeably, it is assumed that both BS and RIS are equipped with extremely large-scale antenna array (e.g., $M, N \geq 256$). For the sake of lower complexity, cost and power consumption [22], [33], the BS is assumed to employ the hybrid beamforming with a limited number of radio frequency (RF) chains, where $N_{\text{RF}} \leq M$ denotes the number of RF chains at the BS. As in the related works [28], [29], [31], [32], it is assumed that the direct channels between the BS and the K users are blocked (see Fig. 1). The channel responses from the k-th user to the RIS and from the RIS to the BS are respectively denoted as $\mathbf{H}_k \in \mathbb{C}^{N\times L}$ and $\mathbf{F} \in \mathbb{C}^{M\times N}$. The reflection vector in the RIS is denoted as

$$\mathbf{v} = [v_1, v_2, \ldots, v_N]^{\mathsf{H}}, \tag{1}$$

with $v_n = e^{j\vartheta_n}$ where $\vartheta_n \in [0, 2\pi)$ denotes the phase shift of the n-th reflective element. We remark that the proposed channel estimation method in Section IV can be applied to both continuous and discrete phase shifting RISs. Given a reflection vector $\mathbf{v}$, the total channel response from the k-th user to the BS is defined as

$$\mathbf{F}\text{diag}(\mathbf{v})\mathbf{H}_k = \mathbf{H}_{[\text{eff},k]} \times (\mathbf{I} \otimes \mathbf{v}), \tag{2}$$

where the *effective* channel, which is independent from the choice of $\mathbf{v}$, is defined as

$$\mathbf{H}_{[\text{eff},k]} \triangleq \left[\mathbf{H}_{[\text{eff},k,1]} \cdots \mathbf{H}_{[\text{eff},k,L]}\right] \in \mathbb{C}^{M\times NL}, \tag{3}$$

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and where the column-wise effective channel is given as

$$\mathbf{H}_{[\text{eff},k,\ell]} = \mathbf{F}\text{diag}\left(\mathbf{H}_k(:,\ell)\right) \in \mathbb{C}^{M\times N},\ \ell \in [L]. \quad (4)$$

Remark 1: Our effective channel in (3) is different from that in [26] and [27]. The effective channel in the latter is expressed as the Kronecker product of the BS-RIS and RIS-User channels, which is defined as a $ML \times N$ matrix. Leveraging this matrix, the joint optimization of a reflection vector and hybrid beamforming matrices requires the computational complexity of $\mathcal{O}\left(M^3L^3\right)$. On the other hand, in our effective channel, the joint optimization can be performed using the column-wise effective channel (4) and the corresponding computational complexity is equal to $\mathcal{O}(M^3)$ [34], [35], [36], [37]. These complexities come from the singular value decomposition (SVD) of the effective or column-wise effective channels. Motivated by this, we in this paper aim at developing an efficient method to estimate the column-wise effective channels $\{\mathbf{H}_{[\text{eff},k,\ell]} : k \in [K], \ell \in [L]\}$ in (4). ■

In the remaining part of this section, we will describe the specific channel models for $\{\mathbf{H}_k : k \in [K]\}$ and $\mathbf{F}$. We first provide the useful definition.

Definition 1: Given a UPA with $N = N_h \times N_v$, a distance between the scatter and the center of array d, an elevation angle $\theta \in [0, 2\pi)$, and an azimuth angle $\phi \in [0, \pi)$, we define a *near-field* array response vector having an element-wise path loss [38]:

$$\mathbf{b}_N(d,\theta,\phi) \triangleq \frac{\mathbf{u}_N(d,\theta,\phi)}{\|\mathbf{u}_N(d,\theta,\phi)\|_2} \in \mathbb{C}^{N\times 1}, \quad (5)$$

where

$$\mathbf{u}_N(d,\theta,\phi) = \left[\frac{e^{-j\frac{2\pi}{\lambda}\Delta d(1,1)}}{\Delta d(1,1)}, \ldots, \frac{e^{-j\frac{2\pi}{\lambda}\Delta d(N_h,N_v)}}{\Delta d(N_h,N_v)}\right]^{\mathsf{T}}.$$

Herein, for each $h \in [N_h]$ and $v \in [N_v]$, we have:

$$\begin{aligned}\Delta d(h,v) = &-\sin\theta\cos\phi x_h - \sin\theta\sin\phi y_v \\ &+ \frac{1-\sin^2\theta\cos^2\phi}{2d}x_h^2 + \frac{1-\sin^2\theta\sin^2\phi}{2d}y_v^2 \\ &- \frac{\sin^2\theta\cos\phi\sin\phi}{d}x_h y_v, \end{aligned} \quad (6)$$

where $x_h = \frac{\lambda}{2}(h-(N_h+1)/2)$ and $y_v = \frac{\lambda}{2}(v-(N_v+1)/2)$. When d is sufficiently large, the last three terms in (5) can be negligible. In this case, the array response vector $\mathbf{b}_N(d,\theta,\phi)$ is simplified as

$$\begin{aligned}\mathbf{a}_N(\theta,\phi) \triangleq \sqrt{\frac{1}{N}}\Big[&1, \ldots, e^{j\pi(h\sin\theta\cos\phi + v\sin\theta\sin\phi)}, \\ &\ldots, e^{j\pi((N_h-1)\sin\theta\cos\phi + (N_v-1)\sin\theta\sin\phi)}\Big]^{\mathsf{T}}. \end{aligned} \quad (7)$$

Note that this is equivalent to the conventional *far-field* UPA array response vector. ■

In Section II-A and Section II-B, we describe the specific channels models according to the distances between the BS and the RIS, and between the RIS and the K users. Throughout the paper, we let z_f and z_{h_k} denote the physical distances between the BS and the RIS, and between the RIS and the user k, respectively.

A. RIS-User Channels

We describe the RIS-User channels, denoted by $\mathbf{H}_k$ for $k \in [K]$. To classify these channels, we define the Rayleigh distance (RD), given by [17]:

$$Z_{\text{RD}} = \frac{2D_{\text{RIS}}^2}{\lambda}, \quad (8)$$

where λ is the wavelength of EM wave and D_{RIS} is the RIS aperture (i.e., $D_{\text{RIS}} = (\lambda/2)\sqrt{N_h^2 + N_v^2}$). As depicted in Fig. 2, each $\mathbf{H}_k$ can be classified into the near- and far-field channels. Specifically, for $z_{h_k} < Z_{\text{RD}}$, $\mathbf{H}_k$ is considered as the near-field channel and the wavefronts are approximated as spherical waves (see Fig. 2 (b)). Otherwise, it is considered as the far-field channel and the wavefronts are approximated as planar waves (see Fig. 2 (a)). From the array response vectors in Definition 1, $\{\mathbf{H}_k : k \in [K]\}$ can be described as follows:

- Near-field channel ($z_{h_k} \le Z_{\text{RD}}$):

$$\begin{aligned}\mathbf{H}_k = \sum_{i=1}^{N_{h_k}} &\alpha_{h_k}^i \mathbf{b}_N\left(d_{[h_k,\text{r}]}^i, \theta_{[h_k,\text{r}]}^i, \phi_{[h_k,\text{r}]}^i\right) \\ &\times \mathbf{a}_L^{\mathsf{H}}\left(\theta_{[h_k,\text{t}]}^i, \phi_{[h_k,\text{t}]}^i\right), \end{aligned} \quad (9)$$

- Far-field channel ($z_{h_k} > Z_{\text{RD}}$):

$$\begin{aligned}\mathbf{H}_k = \sum_{i=1}^{N_{h_k}} &\alpha_{h_k}^i \mathbf{a}_N\left(\theta_{[h_k,\text{r}]}^i \phi_{[h_k,\text{r}]}^i\right) \\ &\times \mathbf{a}_L^{\mathsf{H}}\left(\theta_{[h_k,\text{t}]}^i \phi_{[h_k,\text{t}]}^i\right), \end{aligned} \quad (10)$$

where N_{h_k} is the number of spatial paths, $\alpha_{h_k}^i$ is the channel gain for i-th spatial path, $d_{[h_k,\text{r}]}^i$ is the distance between i-th scatter and the RIS, $\theta_{[h_k,\text{r}]}^i$ (resp. $\theta_{[h_k,\text{t}]}^i$) and $\phi_{[h_k,\text{r}]}^i$ (resp. $\phi_{[h_k,\text{r}]}^i$) are the elevation and azimuth angle of arrival (resp. departure) for the RIS (resp. the user k), respectively.

B. BS-RIS Channel

We describe the BS-RIS channel, denoted by $\mathbf{F}$. To classify this channel, we define the MIMO Rayleigh distance (MIMO-RD), given by [20]:

$$Z_{\text{MRD}} \triangleq \frac{2(D_{\text{RIS}} + D_{\text{BS}})^2}{\lambda}, \quad (11)$$

where $D_{\text{RIS}} = (\lambda/2)\sqrt{N_h^2 + N_v^2}$ and $D_{\text{BS}} = (\lambda/2)\sqrt{M_h^2 + M_v^2}$ denote the antenna apertures of the RIS and the BS, respectively. From the array response vectors in Definition 1, $\mathbf{F}$ can be described as follows:

- Near-field channel ($z_f \le Z_{\text{MRD}}$):

$$\begin{aligned}\mathbf{F} = \sum_{j=1}^{N_f} &\alpha_f^j \mathbf{b}_M\left(d_{[f,\text{r}]}^j, \theta_{[f,\text{r}]}^j, \phi_{[f,\text{r}]}^j\right) \\ &\times \mathbf{b}_N^{\mathsf{H}}\left(d_{[f,\text{t}]}^j, \theta_{[f,\text{t}]}^j, \phi_{[f,\text{t}]}^j\right), \end{aligned} \quad (12)$$

- Far-field channel ($z_f > Z_{\text{MRD}}$):

$$\mathbf{F} = \sum_{j=1}^{N_f} \alpha_f^j \mathbf{a}_M\left(\theta_{[f,\text{r}]}^j, \phi_{[f,\text{r}]}^j\right) \mathbf{a}_N^{\mathsf{H}}\left(\theta_{[f,\text{t}]}^j, \phi_{[f,\text{t}]}^j\right), \quad (13)$$

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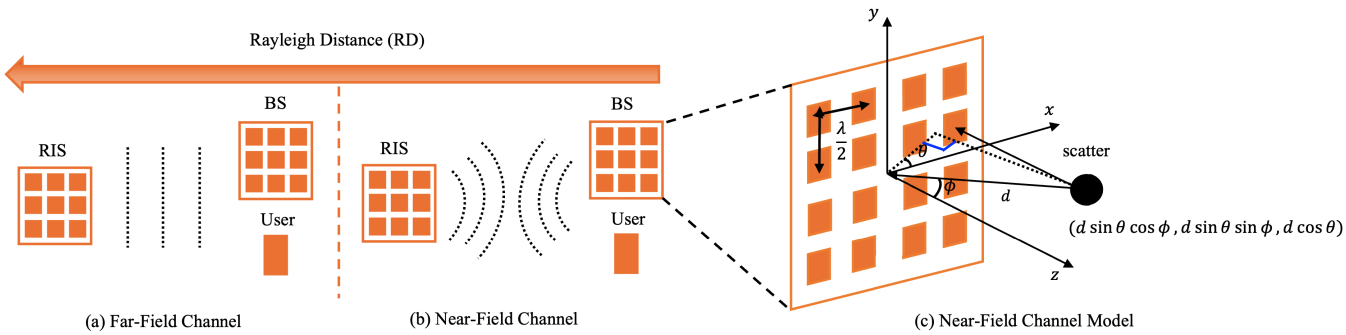

Fig. 2. The categories of wireless channels $\{\mathbf{H}_k : k \in [K]\}$ and $\mathbf{F}$. (a) The far-field region, where the planar-wavefront is assumed. (b) The near-field region, where the spherical-wavefront is assumed. (c) The near-field channel model for UPA communication system.

where N_f is the number of spatial paths, α_f^j is the channel gain for j-th spatial path, $d_{[f,\mathrm{r}]}^j$ (resp. $d_{[f,\mathrm{t}]}^j$) are distance between j-th scatter and the BS (resp. the RIS), $\theta_{[f,\mathrm{r}]}^j$ (resp. $\theta_{[f,\mathrm{t}]}^j$) and $\phi_{[f,\mathrm{r}]}^j$ (resp. $\phi_{[f,\mathrm{t}]}^j$) are elevation and azimuth angle of arrival (resp. departure) for the BS (resp. the RIS), respectively.

For the first time, we propose an *unified* channel estimation method in Section IV, which can perform in the following categories of wireless channels:

1) Far-far field channels: Both $\mathbf{F}$ and $\{\mathbf{H}_k : k \in [K]\}$ are all far-field channels.
2) Far-near field channels: $\mathbf{F}$ is a far-field channel and $\{\mathbf{H}_k : k \in [K]\}$ are all near-field channels.
3) Near-near field channels: Both $\mathbf{F}$ and $\{\mathbf{H}_k : k \in [K]\}$ are all near-field channels.

On the other hand, in the existing compressed sensing (CS)-based methods [13], [31], dictionary matrices should be designed by taking into account the characteristics of the near- and far-field channels.

Remark 2: Note that the BS-RIS and RIS-User channels in Section II-A and Section II-B are represented as the product of the transmitter and receiver far-field or near-field array response vectors. As identified in [20], such decomposition expressions can include the both far-field and near-field non-line-of-sight (NLoS) path components, and far-field line-of-sight (LoS) path components. Also, it was claimed that near-field LoS path components cannot be decoupled by the array response vectors if the distance between the transmitter and the receiver is shorter than a certain value. This value was characterized in [20] only for ULA, which is referred to as MIMO advanced Rayleigh distance (MIMO-ARD). To the best of our knowledge, the MIMO-ARD for UPA, which is our system model, has not been characterized yet. Regarding the near-field channels, the proposed method can be applied to the entire near-field NLoS channels and some near-field LoS channels. For the other near-field LoS channels (i.e., non-decoupled channels), there is no prior work on the channel estimation methods. This is beyond the scope of this work and is left for a future work. ■

III. Channel Estimation Protocol

The channel estimation protocol performs with J subframes, each of which consists of T symbols (or time slots) with $T \geq KL$. Accordingly, it requires the overall JT symbols, where the hyperparameter J can control the tradeoff between the estimation accuracy and the training overhead. Throughout the paper, J is referred to as the training overhead. In the proposed protocol, the RIS reflection vector $\mathbf{v}_j = [v_{[j,1]}, v_{[j,2]}, \ldots, v_{[j,N]}]^\intercal$ and the $M \times N_{\mathrm{RF}}$ RF combining matrix $\mathbf{C}_j$ are unchanged within each subframe $j \in [J]$, but they can be changed across the subframes. Their specific constructions will be provided in Section IV.

During the J subframes, our channel estimation protocol proceeds as follows. At every subframe $j \in [J]$, each user $k \in [K]$ transmits its orthogonal pilot sequence of the length T to the BS, denoted by $\mathbf{X}_{[j,k]} \in \mathbb{C}^{L\times T}$, such that

$$\mathbf{X}_{[j,k]}\mathbf{X}_{[j,k]}^{\mathsf{H}} = PT \times \mathbf{I}, \tag{14}$$

where PT is the energy constraint, P is the transmit power of each user and antenna, and $\mathbf{X}_{[j,k]}\mathbf{X}_{[j,k']}^{\mathsf{H}} = \mathbf{0}$ if $k \neq k'$. This construction is always possible due to the choice of $T \geq KL$. Note that each user k sends the i-th row of $\mathbf{X}_{[j,k]}$ through the i-th transmit antenna during the T time slots. For each subframe $j \in [J]$, the BS observes the $N_{\mathrm{RF}} \times T$ matrix, given by

$$\mathbf{Y}_j = \mathbf{C}_j^{\mathsf{H}}\left(\sum_{k=1}^{K}\left(\mathbf{F}\mathrm{diag}(\mathbf{v}_j)\mathbf{H}_k\right)\mathbf{X}_{[j,k]} + \mathbf{U}_j\right), \tag{15}$$

where $\mathbf{U}_j$ is the noise matrix whose elements follow independently identically circularly symmetric complex Gaussian distribution with mean zero and variance σ^2. From the orthogonality of the pilot sequences, we can derive the user-specific observation:

$$\begin{aligned}\mathbf{Z}_{[j,k]} &= \frac{1}{PT}\mathbf{Y}_j\mathbf{X}_{[j,k]}^{\mathsf{H}} \in \mathbb{C}^{N_{\mathrm{RF}}\times L}\\ &= \mathbf{C}_j^{\mathsf{H}}\left(\mathbf{F}\mathrm{diag}(\mathbf{v}_j)\mathbf{H}_k + \mathbf{N}_{[j,k]}\right),\end{aligned} \tag{16}$$

for $k \in [K]$, where $\mathbf{N}_{[j,k]} \triangleq \frac{1}{PT}\mathbf{U}_j\mathbf{X}_{[j,k]}^{\mathsf{H}}$. Stacking the K user-specific observations, we define:

$$\begin{aligned}\mathbf{Z}_j &\triangleq \left[\mathbf{Z}_{[j,1]} \cdots \mathbf{Z}_{[j,K]}\right]\\ &= \mathbf{C}_j^{\mathsf{H}}\left(\mathbf{F}\mathrm{diag}(\mathbf{v}_j)\mathbf{H} + \mathbf{N}_j\right),\ j \in [J],\end{aligned} \tag{17}$$

where the stacked channel $\mathbf{H}$ and noise matrix $\mathbf{N}_j$ are respectively defined as

$$\mathbf{H} \triangleq \left[\mathbf{H}_1 \cdots \mathbf{H}_K\right] \in \mathbb{C}^{N\times KL}$$

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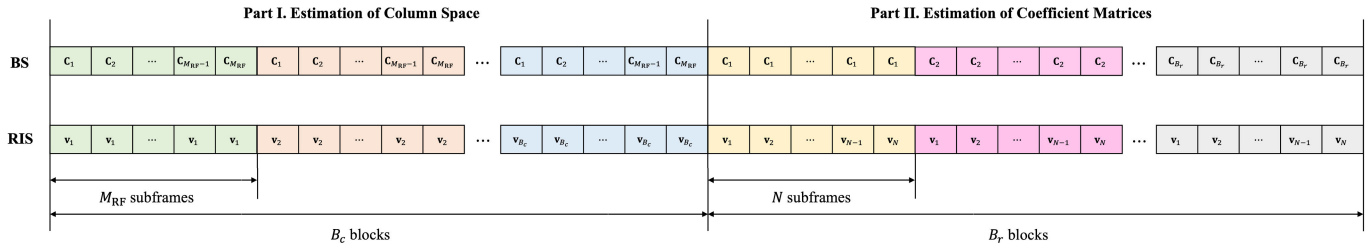

Fig. 3. The proposed channel estimation protocol and frame structure, where each subframe consists of T (i.e., the length of a pilot sequence) time slots, and $M_{\mathrm{RF}} = M/N_{\mathrm{RF}}$.

$$\mathbf{N}_j \triangleq \left[\mathbf{N}_{[j,1]} \cdots \mathbf{N}_{[j,K]}\right] \in \mathbb{C}^{N\times KL}.$$

Remark 3: Due to the use of hybrid beamforming structures, the BS can observe the $N_{\mathrm{RF}} \times L$ matrix during each subframe. On the other hand, when (conventional) fully-digital beamforming is employed, the $M \times L$ matrix is observed. To attain the same amount of the training observations, the training overhead becomes the M/N_{RF} times bigger. Naturally, in our simulations in Section V, the scale of the training overhead tends to be larger than those in [12], [13], and [39] based on fully-digital systems. ■

IV. The Proposed Channel Estimation Method

In this section, we present an efficient method to estimate the effective channels $\{\mathbf{H}_{[\mathrm{eff},k]} : k \in [K]\}$ in (3) from the processed training observations $\{\mathbf{Z}_j : j \in [J]\}$ in (17). As shown in Section II-B, the BS-RIS channel (i.e., $\mathbf{F}$) is expressed as the linear combination of the N_f number of rank-1 matrices in the both *near-* and *far-field* channels. Since the rank of $\mathbf{F}$ (denoted by $\mathrm{rk}(\mathbf{F})$) is less than or equal to N_f, $\mathbf{F}$ is a low-rank matrix in mmWave channels. This is one of the key ingredients for the proposed channel estimation method. Let $\mathbf{S}_{\mathrm{col}}$ denote the $M \times \mathrm{rk}(\mathbf{F})$ matrix whose columns are the basis of the column space of $\mathbf{F}$ (a.k.a. the common column space). Regardless of the channel characteristics (i.e., the near- and far-field channels) of $\mathbf{F}$, each column-wise effective channel $\mathbf{H}_{[\mathrm{eff},k,\ell]}$ in (4) can be represented as the *low-rank* decomposition:

$$\mathbf{H}_{[\mathrm{eff},k,\ell]} = \mathbf{S}_{\mathrm{col}}\mathbf{T}_{[k,\ell]},\ k \in [K], \ell \in [L], \tag{18}$$

where $\mathbf{T}_{[k,\ell]} \in \mathbb{C}^{\mathrm{rk}(\mathbf{F})\times N}$ denotes a user and antenna specific coefficient matrix. On the basis of the low-rank decomposition, the proposed channel estimation method consists of the two parts (see Fig. 3). In the first part, using the training observations from the first $M_{\mathrm{RF}}B$ subframes, the column space of $\mathbf{F}$ (i.e., $\mathbf{S}_{\mathrm{col}}$) is estimated by means of a collaborative low-rank approximation. Then, using the training observations from the remaining NB_r subframes, the coefficient matrices $\mathbf{T}_{[k,\ell]}$'s are jointly optimized. The proposed channel estimation method is named **C**ollaborative **L**ow-**R**ank **A**pproximation and **J**oint **O**ptimization (CLRA-JO). Also, the training overhead of the proposed CLRA-JO is computed as

$$J = M_{\mathrm{RF}}B_c + NB_r, \tag{19}$$

where B_c and B_r denote the hyperparameters which can control the tradeoff between the estimation accuracy and the training overhead.

In the subsequent subsections, we will design the RF combining matrices $\{\mathbf{C}_j : j \in [J]\}$ and the reflection vectors $\{\mathbf{v}_j : j \in [J]\}$, in order to derive the processed observations suitable for the proposed CLRA-JO. Before describing them, we provide the useful definition:

Definition 2: Let $\mathbf{\Phi}_{[X,Y]}$ be a $X \times Y$ matrix consisting of orthonormal columns. In the following, we will construct the RF combining matrices and the reflection vectors using the matrix $\mathbf{\Phi}_{[X,Y]}$. In practice, a DFT matrix can be adopted as the $\mathbf{\Phi}_{[X,Y]}$. ■

A. Estimation of Column Space

In the first part, the overall $M_{\mathrm{RF}}B_c$ subframes are used to estimate $\mathbf{S}_{\mathrm{col}}$ (see Fig. 3). To design the RF combining matrices and the reflection vectors, the subframes are partitioned into B_c blocks each having M_{RF} subframes.

From the training observations $\{\mathbf{Z}_j : j \in [M_{\mathrm{RF}}B_c]\}$ in (17), we derive the observations suitable to estimate $\mathbf{S}_{\mathrm{col}}$ by properly designing $\{\mathbf{C}_j, \mathbf{v}_j : j \in [M_{\mathrm{RF}}B_c]\}$. For the compactness of expressions, we re-index the training observations $\{\mathbf{Z}_j : j \in [M_{\mathrm{RF}}B_c]\}$ such as

$$\mathbf{Z}^{\mathrm{1st}}_{[i,b]} \triangleq \mathbf{Z}_{M_{\mathrm{RF}}(b-1)+i}, \tag{20}$$

for $i \in [M_{\mathrm{RF}}]$ and $b \in [B_c]$. The RF combining matrices are designed as

$$\mathbf{C}_{M_{\mathrm{RF}}(b-1)+i} = \mathbf{\Phi}_{[M,M]}(:, [N_{\mathrm{RF}}(i-1)+1 : N_{\mathrm{RF}}i]), \tag{21}$$

for each $i \in [M_{\mathrm{RF}}]$ and for every $b \in [B_c]$. As shown in Fig. 3, $\mathbf{C}_i$ is used at the i-th subframe of every block, namely, the subframe-dependent RF-combining matrices are used. For each block $b \in [B_c]$, the M_{RF} number of reflection vectors are constructed as follows:

$$\mathbf{v}_{M_{\mathrm{RF}}(b-1)+i} = \mathbf{\Phi}_{[N,B_c]}(:, b),\ \forall i \in [M_{\mathrm{RF}}].$$

Focusing on the block $b \in [B_c]$, we will explain how to obtain the column-sampled observations to estimate $\mathbf{S}_{\mathrm{col}}$. The exactly same procedures are applied to every block. Letting $\mathbf{Q}_b \triangleq \mathrm{diag}(\mathbf{v}_{M_{\mathrm{RF}}(b-1)+i})\mathbf{H} = \mathrm{diag}(\mathbf{\Phi}_{[N,B_c]}(:, b))\mathbf{H}$ and with the above constructions, we obtain the following observations from (17):

$$\mathbf{Z}^{\mathrm{1st}}_{[i,b]} = \mathbf{C}^{\mathsf{H}}_{M_{\mathrm{RF}}(b-1)+i}\mathbf{F}\mathbf{Q}_b + \tilde{\mathbf{N}}_{[i,b]}, \tag{22}$$

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where $\tilde{\mathbf{N}}_{[i,b]} = \mathbf{C}^{\mathsf{H}}_{M_{\mathrm{RF}}(b-1)+i}\mathbf{N}_{M_{\mathrm{RF}}(b-1)+i}$. For each block $b \in [B_c]$, we can get:

$$\begin{aligned}\tilde{\mathbf{Z}}_{[\mathrm{col},b]} &= \mathbf{\Phi}_{[M,M]}\begin{bmatrix}\mathbf{Z}^{\mathrm{1st}}_{[1,b]}\\ \cdot\\ \cdot\\ \cdot\\ \mathbf{Z}^{\mathrm{1st}}_{[M_{\mathrm{RF}},b]}\end{bmatrix} \in \mathbb{C}^{M\times KL}\\ &\overset{(a)}{=} \mathbf{\Phi}_{[M,M]}\mathbf{\Phi}^{\mathsf{H}}_{[M,M]}\left(\mathbf{F}\mathbf{Q}_b + \tilde{\mathbf{N}}_{[\mathrm{col},b]}\right)\\ &= \mathbf{F}\mathbf{Q}_b + \tilde{\mathbf{N}}_{[\mathrm{col},b]},\end{aligned} \tag{23}$$

where (a) is due to the structure of the RF combining matrices in (21) and

$$\tilde{\mathbf{N}}_{[\mathrm{col},b]} = \mathbf{\Phi}_{[M,M]}\begin{bmatrix}\tilde{\mathbf{N}}_{[1,b]}\\ \cdot\\ \cdot\\ \cdot\\ \tilde{\mathbf{N}}_{[M_{\mathrm{RF}},b]}\end{bmatrix}.$$

From $\{\tilde{\mathbf{Z}}_{[\mathrm{col},b]} : b \in [B_c]\}$, we can obtain the processed observations suitable for estimating $\mathbf{S}_{\mathrm{col}}$:

$$\begin{aligned}\mathbf{M}_{\mathrm{col}} &\triangleq \left[\tilde{\mathbf{Z}}_{[\mathrm{col},1]} \cdots \tilde{\mathbf{Z}}_{[\mathrm{col},B_c]}\right] \in \mathbb{C}^{M\times B_cKL}\\ &= \mathbf{F}\mathbf{Q} + \tilde{\mathbf{N}}_{\mathrm{col}},\end{aligned} \tag{24}$$

where $\tilde{\mathbf{N}}_{\mathrm{col}} \triangleq \left[\tilde{\mathbf{N}}_{[\mathrm{col},1]} \ldots \tilde{\mathbf{N}}_{[\mathrm{col},B_c]}\right]$ and $\mathbf{Q} \triangleq [\mathbf{Q}_1 \ldots \mathbf{Q}_{B_c}]$. Noticeably, since $\mathbf{Q}$ is a full-rank matrix with high-probability (i.e., $\mathrm{rank}(\mathbf{Q}) = \mathrm{rank}(\mathbf{H}) = N$ with high-probability), the column space of $\mathbf{M}_{\mathrm{col}}$ can be equivalent to that of $\mathbf{F}$, provided that the impact of noise is very small.

We explain how to estimate $\mathbf{S}_{\mathrm{col}}$ from the processed observations $\mathbf{M}_{\mathrm{col}}$ in (24). Recall that $\mathbf{S}_{\mathrm{col}}$ is the $M \times \mathrm{rk}(\mathbf{F})$ matrix whose columns are the basis of the column space of $\mathbf{F}$. To estimate it, we find the eigenvalue decomposition of the covariance matrix of $\mathbf{M}_{\mathrm{col}}$:

$$\mathbf{M}_{\mathrm{col}}\mathbf{M}^{\mathsf{H}}_{\mathrm{col}} = \tilde{\mathbf{S}}_{\mathrm{col}}\mathbf{\Sigma}_{\mathrm{col}}\tilde{\mathbf{S}}^{\mathsf{H}}_{\mathrm{col}}, \tag{25}$$

where $\tilde{\mathbf{S}}_{\mathrm{col}}$ is the eigenvectors corresponding to the eigenvalue matrix

$$\mathbf{\Sigma}_{\mathrm{col}} = \mathrm{diag}\left([\lambda_{[\mathrm{col},1]}, \lambda_{[\mathrm{col},2]}, \ldots, \lambda_{[\mathrm{col},M]}]\right),$$

where the eigenvalues $\lambda_{[\mathrm{col},\ell]}$'s are ordered in a descending order in magnitude. As suggested in [13], the $\mathrm{rk}(\mathbf{F})$ (i.e., the rank of $\mathbf{F}$ (or $\mathbf{S}_{\mathrm{col}}$)) can be efficiently estimated using the minimum description length (MDL) criterion [40] as

$$\begin{aligned}&\widehat{\mathrm{rk}}(\mathbf{F})\\ &= \arg\min_{n\in[M]}\left\{-\log\left(\frac{\prod_{i=n+1}^{M}\lambda_{[\mathrm{col},i]}^{\frac{1}{M-n}}}{\frac{1}{M-n}\sum_{i=n+1}^{M}\lambda_{[\mathrm{col},i]}}\right)^{(M-n)B_cT}\right.\\ &\quad\left.+\frac{1}{2}n(2M-n)\log(B_cT)\right\}.\end{aligned} \tag{26}$$

As long as independent columns are sufficiently sampled, the optimization below can yield a good $\widehat{\mathrm{rk}}(\mathbf{F})$-dimensional column space of $\mathbf{F}$ [41]:

$$\hat{\mathbf{S}}_{\mathrm{col}} = \underset{\mathbf{S}\in\mathbb{C}^{M\times\widehat{\mathrm{rk}}(\mathbf{F})},\ \mathbf{S}^{\mathsf{H}}\mathbf{S}=\mathbf{I}}{\arg\min}\left\|\left(\mathbf{I}-\mathbf{S}\mathbf{S}^{\mathsf{H}}\right)\mathbf{M}_{\mathrm{col}}\mathbf{M}^{\mathsf{H}}_{\mathrm{col}}\right\|_2^2. \tag{27}$$

The objective function can measure the distance between the column spaces of $\mathbf{S}$ and $\mathbf{M}_{\mathrm{col}}$ since the difference of $\mathbf{S}\mathbf{S}^{\mathsf{H}}\mathbf{M}_{\mathrm{col}}$ (i.e., the projections of the columns of $\mathbf{M}_{\mathrm{col}}$ onto the column space of $\mathbf{S}$) and $\mathbf{M}_{\mathrm{col}}$ should be zero if $\mathbf{S}$ and $\mathbf{M}_{\mathrm{col}}$ has the same column space. From Eckart-Young-Mirsky Theorem [42], the optimal solution to the above problem is derived as

$$\hat{\mathbf{S}}_{\mathrm{col}} = \tilde{\mathbf{S}}_{\mathrm{col}}(:, [\widehat{\mathrm{rk}}(\mathbf{F})]). \tag{28}$$

As in [43], the accuracy of the estimated column space can be measured by the Euclidean distance between the BS-RIS channel (i.e., $\mathbf{F}$) and the BS-RIS channel projected on the estimated column space (i.e., $\hat{\mathbf{S}}_{\mathrm{col}}\hat{\mathbf{S}}^{\mathsf{H}}_{\mathrm{col}}\mathbf{F}$). Also, its upper bound is derived as

$$\left\|\left(\mathbf{I}-\hat{\mathbf{S}}_{\mathrm{col}}\hat{\mathbf{S}}^{\mathsf{H}}_{\mathrm{col}}\right)\mathbf{F}\right\|_F^2 \le \frac{\delta}{B_cKL}, \tag{29}$$

for some $\delta > 0$. We can obtain a *multi-user gain* since the upper bound can decrease as K grows. Namely, given a target accuracy level, we can reduce the pilot overhead B_c inversely proportional to K.

B. Estimation of Coefficient Matrices

We propose an iterative algorithm to estimate the coefficient matrices $\{\mathbf{T}_{[k,\ell]} : k \in [K], \ell \in [L]\}$ in (18). We first identify the some relationship among the column-wise effective channels due to their special structure in (4):

$$\begin{aligned}\mathbf{H}_{[\mathrm{eff},k,\ell]} &= \mathbf{F}\mathrm{diag}(\mathbf{H}_k(:,\ell))\\ &= \mathbf{F}\left[\mathrm{diag}(\mathbf{H}_1(:,1))\mathrm{diag}(\mathbf{H}_1(:,1))^{-1}\right]\mathrm{diag}(\mathbf{H}_k(:,\ell))\\ &= \mathbf{F}\mathrm{diag}(\mathbf{H}_1(:,1))\mathbf{D}_{[k,\ell]} = \mathbf{H}_{[\mathrm{eff},1,1]}\mathbf{D}_{[k,\ell]},\end{aligned}$$

where $\mathbf{D}_{[k,\ell]} \triangleq \mathrm{diag}(\mathbf{H}_1(:,1))^{-1}\mathrm{diag}(\mathbf{H}_k(:,\ell))$ is the $N \times N$ diagonal matrix. Since $\mathbf{H}_{[\mathrm{eff},1,1]} = \mathbf{S}_{\mathrm{col}}\mathbf{T}_{[1,1]}$, each coefficient matrix $\mathbf{T}_{[k,\ell]}$ can be expressed as

$$\mathbf{T}_{[k,\ell]} = \mathbf{T}_{[1,1]}\mathbf{D}_{[k,\ell]},\ k \in [K], \ell \in [L]. \tag{30}$$

We need to estimate $\mathbf{T}_{[1,1]}$ and the diagonal matrices $\{\mathbf{D}_{[k,\ell]} : k \in [K], \ell \in [L]\}$. To design RF combining matrices and reflection vectors, the subframes in the second part are partitioned into B_r blocks each having N subframes (see Fig. 3). For ease of exposition, we let $n_0 \triangleq M_{\mathrm{RF}}B$ denote the index of the last subframe in the first part. From the training observations $\{\mathbf{Z}_j : j \in [n_0+1 : J]\}$ in (17), we derive the processed observations suitable to estimate $\mathbf{T}_{[1,1]}$ and $\{\mathbf{D}_{[k,\ell]} : k \in [K], \ell \in [L]\}$ by properly designing $\{\mathbf{C}_j, \mathbf{v}_j : j \in [n_0+1 : J]\}$. As before, we re-index the training observations $\{\mathbf{Z}_j : j \in [n_0+1 : J]\}$ such as

$$\mathbf{Z}^{\mathrm{2nd}}_{[i,b]} \triangleq \mathbf{Z}_{n_0+M_{\mathrm{RF}}(b-1)+i},\ b \in [B_r], i \in [N]. \tag{31}$$

For each $b \in [B_r]$ and for every $i \in [N]$, the RF combining matrices $\mathbf{C}_{n_0+M_{\mathrm{RF}}(b-1)+i}$ are designed as

$$\mathbf{\Phi}_{[M,N_{\mathrm{RF}}B_r]}(:, [N_{\mathrm{RF}}(b-1)+1 : N_{\mathrm{RF}}b]). \tag{32}$$

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For each $i \in [N]$ and for every $b \in [B_r]$, the reflection vector is constructed as

$$\mathbf{v}_{n_0+M_{\mathrm{RF}}(b-1)+i} = \mathbf{\Phi}_{[N,N]}(:,i). \tag{33}$$

With the above constructions and from (17) and (31), we can define an user- and antenna-wise observation:

$$\begin{aligned}\mathbf{Z}^{\mathrm{2nd}}_{[i,b,k,\ell]} &\triangleq \mathbf{Z}^{\mathrm{2nd}}_{[i,b]}(:, L(k-1)+\ell)\\ &\overset{(a)}{=} \mathbf{C}^{\mathsf{H}}_{n_0+M_{\mathrm{RF}}(b-1)+i}\mathbf{H}_{[\mathrm{eff},k,\ell]}\mathbf{\Phi}_{[N,N]}(:,i) + \tilde{\mathbf{N}}_{[i,b,k,\ell]},\end{aligned}$$

where (a) follows from (4), (16), and (33), and

$$\tilde{\mathbf{N}}_{[i,b,k,\ell]} = \mathbf{C}^{\mathsf{H}}_{n_0+M_{\mathrm{RF}}(b-1)+i}\mathbf{N}_{n_0+M_{\mathrm{RF}}(b-1)+i}(:, L(k-1)+\ell).$$

For each block $b \in [B_r]$, we can get:

$$\begin{aligned}\bar{\mathbf{Z}}_{[b,k,\ell]} &\triangleq \left[\mathbf{Z}^{\mathrm{2nd}}_{[1,b,k,\ell]} \cdots \mathbf{Z}^{\mathrm{2nd}}_{[N,b,k,\ell]}\right]\mathbf{\Phi}^{\mathsf{H}}_{[N,N]}\\ &= \mathbf{C}^{\mathsf{H}}_{n_0+M_{\mathrm{RF}}(b-1)+i}\mathbf{H}_{[\mathrm{eff},k,\ell]}\mathbf{\Phi}_{[N,N]}\mathbf{\Phi}^{\mathsf{H}}_{[N,N]} + \tilde{\mathbf{N}}_{[b,k,\ell]}\\ &= \mathbf{C}^{\mathsf{H}}_{n_0+M_{\mathrm{RF}}(b-1)+i}\mathbf{H}_{[\mathrm{eff},k,\ell]} + \tilde{\mathbf{N}}_{[b,k,\ell]},\end{aligned}$$

where

$$\tilde{\mathbf{N}}_{[b,k,\ell]} = \left[\tilde{\mathbf{N}}_{[1,b,k,\ell]} \ldots \tilde{\mathbf{N}}_{[N,b,k,\ell]}\right]\mathbf{\Phi}^{\mathsf{H}}_{[N,N]}.$$

Stacking $\{\bar{\mathbf{Z}}_{[b,k,\ell]} : b \in [B_r]\}$, we define:

$$\begin{aligned}\mathbf{M}_{[\mathrm{row},k,\ell]} &\triangleq \begin{bmatrix}\bar{\mathbf{Z}}_{[1,k,\ell]}\\ \cdot\\ \cdot\\ \cdot\\ \bar{\mathbf{Z}}_{[B_r,k,\ell]}\end{bmatrix} \in \mathbb{C}^{N_{\mathrm{RF}}B_r \times N}\\ &\overset{(a)}{=} \mathbf{\Phi}^{\mathsf{H}}_{[M,N_{\mathrm{RF}}B_r]}\mathbf{H}_{[\mathrm{eff},k,\ell]} + \tilde{\mathbf{N}}_{[\mathrm{row},k,\ell]}\\ &\overset{(b)}{=} \mathbf{\Phi}^{\mathsf{H}}_{[M,N_{\mathrm{RF}}B_r]}\mathbf{S}_{\mathrm{col}}\mathbf{T}_{[k,\ell]} + \tilde{\mathbf{N}}_{[\mathrm{row},k,\ell]},\end{aligned} \tag{34}$$

where (a) is due to the proposed RF combining matrices in (32), (b) is from the decomposition in (18), and

$$\tilde{\mathbf{N}}_{[\mathrm{row},k,\ell]} \triangleq \begin{bmatrix}\tilde{\mathbf{N}}_{[1,k,\ell]}\\ \cdot\\ \cdot\\ \cdot\\ \tilde{\mathbf{N}}_{[B_r,k,\ell]}\end{bmatrix}.$$

Letting $\mathbf{P} \triangleq \mathbf{\Phi}^{\mathsf{H}}_{[M,N_{\mathrm{RF}}B_r]}\hat{\mathbf{S}}_{\mathrm{col}} \in \mathbb{C}^{N_{\mathrm{RF}}B_r \times \widehat{\mathrm{rk}}(\mathbf{F})}$, we can derive the solution of the individual least-square (LS) estimation using the pseudo-inverse $\mathbf{P}^{\dagger}$:

$$\begin{aligned}\hat{\mathbf{T}}^{\mathrm{LS}}_{[k,\ell]} &\triangleq \mathbf{P}^{\dagger}\mathbf{M}_{[\mathrm{row},k,\ell]} \in \mathbb{C}^{\widehat{\mathrm{rk}}(\mathbf{F})\times N}\\ &= \mathbf{P}^{\dagger}\left(\mathbf{\Phi}^{\mathsf{H}}_{[M,N_{\mathrm{RF}}B_r]}\mathbf{H}_{[\mathrm{eff},k,\ell]} + \tilde{\mathbf{N}}_{[\mathrm{row},k,\ell]}\right)\\ &= \mathbf{P}^{\dagger}\left(\mathbf{\Phi}^{\mathsf{H}}_{[M,N_{\mathrm{RF}}B_r]}\mathbf{S}_{\mathrm{col}}\mathbf{T}_{[k,\ell]} + \tilde{\mathbf{N}}_{[\mathrm{row},k,\ell]}\right).\end{aligned} \tag{35}$$

The existence of the $\mathbf{P}^{\dagger}$ can be guaranteed by choosing the hyperparameter B_r such that

$$N_{\mathrm{RF}}B_r \geq \widehat{\mathrm{rk}}(\mathbf{F}). \tag{36}$$

Normally, we have $B_r = 1$ for very sparse channel, i.e., $N_{\mathrm{RF}} \geq \widehat{\mathrm{rk}}(\mathbf{F})$.

The LS solutions would be good in very high SNRs. However, it is well-known that the multiplication of the pseudo-inverse would increase the noise power, thereby degrading the performance in practical SNRs. We address this limitation by harnessing the special property in (30). Toward this, we jointly optimize the coefficient matrices by taking the solutions of

$$\begin{aligned}&\underset{\mathbf{T}_{[1,1]},\mathbf{D}_{[1,1]},\ldots,\mathbf{D}_{[K,L]}}{\arg\min} \mathcal{L}\left(\mathbf{T}_{[1,1]}, \{\mathbf{D}_{[k,\ell]}\}\right)\\ &\quad \text{subject to } \{\mathbf{D}_{[k,\ell]}\} \text{ are diagonal matrices,}\end{aligned} \tag{37}$$

where

$$\mathcal{L}\left(\mathbf{T}_{[1,1]}, \{\mathbf{D}_{[k,\ell]}\}\right) \triangleq \sum_{k=1}^{K}\sum_{\ell=1}^{L}\left\|\hat{\mathbf{T}}^{\mathrm{LS}}_{[k,\ell]} - \mathbf{T}_{[1,1]}\mathbf{D}_{[k,\ell]}\right\|^2_F.$$

We efficiently solve this problem via an alternating optimization (see Algorithm 1).

Iterations. At the t-th iteration, the proposed alternating optimization is performed as follows:

- For the fixed $\hat{\mathbf{T}}^{(t-1)}_{[1,1]}$, our optimization can be formulated as

$$\begin{aligned}\hat{\mathbf{D}}^{(t)}_{[k,\ell]} &= \underset{\mathbf{D}_{[k,\ell]}}{\arg\min}\, \mathcal{L}\left(\hat{\mathbf{T}}^{(t-1)}_{[1,1]}, \mathbf{D}_{[k,\ell]} \middle| \hat{\mathbf{T}}^{(t-1)}_{[1,1]}\right)\\ &= \underset{\mathbf{D}_{[k,\ell]}}{\arg\min}\left\|\hat{\mathbf{T}}^{\mathrm{LS}}_{[k,\ell]} - \hat{\mathbf{T}}^{(t-1)}_{[1,1]}\mathbf{D}_{[k,\ell]}\right\|^2_F,\end{aligned} \tag{38}$$

  for $k \in [K]$, $\ell \in [L]$, where the initial value is determined as $\hat{\mathbf{T}}^{(0)}_{[1,1]} = \hat{\mathbf{T}}^{\mathrm{LS}}_{[1,1]}$. This LS problem is easily solved as

$$\hat{\mathbf{D}}^{(t)}_{[k,\ell]} = \mathrm{diag}\left(\mathbf{R}(:,\Omega_N)^{\dagger}\hat{\mathbf{t}}^{\mathrm{LS}}_{[k,\ell]}(\Omega_N)\right), \tag{39}$$

  where $\mathbf{R} = \mathbf{I} \otimes \hat{\mathbf{T}}^{(t-1)}_{[1,1]} \in \mathbb{C}^{N_f N \times N^2}$ and $\Omega_N = \{n + (n-1)N : n \in [N]\}$. Also, the vectorization of $\hat{\mathbf{T}}^{\mathrm{LS}}_{[k,\ell]}$ is denoted as $\hat{\mathbf{t}}^{\mathrm{LS}}_{[k,\ell]} = \mathrm{vec}(\hat{\mathbf{T}}^{\mathrm{LS}}_{[k,\ell]}) \in \mathbb{C}^{\widehat{\mathrm{rk}}(\mathbf{F})N\times 1}$.
- For the fixed $\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]} : k \in [K], \ell \in [L]\}$, our optimization can be formulated as the standard LS problem:

$$\begin{aligned}\hat{\mathbf{T}}^{(t)}_{[1,1]} &= \underset{\mathbf{T}}{\arg\min}\, \mathcal{L}\left(\mathbf{T}, \{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\} \middle| \{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\}\right)\\ &= \underset{\mathbf{T}}{\arg\min}\sum_{k=1}^{K}\sum_{\ell=1}^{L}\left\|\hat{\mathbf{T}}^{\mathrm{LS}}_{[k,\ell]} - \mathbf{T}\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\|^2_F,\end{aligned} \tag{40}$$

  where $\hat{\mathbf{D}}^{(t)}_{[1,1]} = \mathbf{I}$ during iterations. The optimal solution is derived as

$$\begin{aligned}\hat{\mathbf{T}}^{(t)}_{[1,1]} = &\left(\sum_{k=1}^{K}\sum_{\ell=1}^{L}\hat{\mathbf{T}}^{\mathrm{LS}}_{[k,\ell]}\left(\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right)^{\mathsf{H}}\right)\\ &\times\left(\sum_{k=1}^{K}\sum_{\ell=1}^{L}\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\left(\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right)^{\mathsf{H}}\right)^{-1}.\end{aligned} \tag{41}$$

After the $t_{\max}$ iterations, the estimated column-wise effective channels are obtained as

$$\hat{\mathbf{H}}_{[\mathrm{eff},k,\ell]} = \hat{\mathbf{S}}_{\mathrm{col}}\hat{\mathbf{T}}^{(t_{\max})}_{[1,1]}\hat{\mathbf{D}}^{(t_{\max})}_{[k,\ell]}. \tag{42}$$

C. Analysis

In this section, we analyze the convergence and the computational complexity of the proposed channel estimation method.

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Algorithm 1 Proposed CLRA-JO Algorithm

1: **Input:** $\hat{\mathbf{S}}_{\text{col}}$ (i.e., the estimated column space of $\mathbf{F}$) in (28), $\{\mathbf{M}_{[\text{row},k,\ell]} : k \in [K], \ell \in [L]\}$ in (34), and the maximum number of iterations $t_{\max}$.

2: **Initialization:** Set $\hat{\mathbf{T}}^{(0)}_{[1,1]} = \hat{\mathbf{T}}^{\text{LS}}_{[1,1]}$ from (35).

3: **Repeat until** $t = t_{\max}$

- Given $\hat{\mathbf{T}}^{(t-1)}_{[1,1]}$, update the $\hat{\mathbf{D}}^{(t)}_{[k,\ell]}$ via (39).
- Given $\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]} : k \in [K], \ell \in [L]\}$, update $\hat{\mathbf{T}}^{(t)}_{[1,1]}$ via (41).

4: **Output:** The estimated RIS-aided effective channels:

$$\hat{\mathbf{H}}_{[\text{eff},k,\ell]} = \hat{\mathbf{S}}_{\text{col}} \hat{\mathbf{T}}^{(t_{\max})}_{[1,1]} \hat{\mathbf{D}}^{(t_{\max})}_{[k,\ell]}.$$

1) Convergence Analysis: We prove the convergence of the proposed iterative algorithm in Algorithm 1. Toward this, we show that

$$\mathcal{L}\left(\hat{\mathbf{T}}^{(t-1)}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t-1)}_{[k,\ell]}\right\}\right) \geq \mathcal{L}\left(\hat{\mathbf{T}}^{(t)}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\}\right). \tag{43}$$

This implies that as t grows, the $\hat{\mathbf{T}}^{(t)}_{[1,1]}$ and $\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\}$ converge to the optimal solution of our optimization in (37). Specifically, the proof is provided as follows:

$$\begin{aligned}
&\mathcal{L}\left(\hat{\mathbf{T}}^{(t-1)}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t-1)}_{[k,\ell]}\right\}\right) \\
&= \mathcal{L}\left(\hat{\mathbf{T}}^{(t-1)}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t-1)}_{[k,\ell]}\right\} \middle| \hat{\mathbf{T}}^{(t-1)}_{[1,1]}\right) \\
&\overset{(a)}{\geq} \min_{\{\mathbf{D}[k,\ell]\}} \mathcal{L}\left(\hat{\mathbf{T}}^{(t-1)}_{[1,1]}, \{\mathbf{D}_{[k,\ell]}\} \middle| \hat{\mathbf{T}}^{(t-1)}_{[1,1]}\right) \\
&\overset{(b)}{=} \mathcal{L}\left(\hat{\mathbf{T}}^{(t-1)}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\} \middle| \hat{\mathbf{T}}^{(t-1)}_{[1,1]}\right) \\
&= \mathcal{L}\left(\hat{\mathbf{T}}^{(t-1)}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\}\right) \\
&= \mathcal{L}\left(\hat{\mathbf{T}}^{(t-1)}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\} \middle| \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\}\right) \\
&\overset{(c)}{\geq} \min_{\mathbf{T}[1,1]} \mathcal{L}\left(\mathbf{T}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\} \middle| \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\}\right) \\
&\overset{(d)}{=} \mathcal{L}\left(\hat{\mathbf{T}}^{(t)}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\} \middle| \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\}\right) \\
&= \mathcal{L}\left(\hat{\mathbf{T}}^{(t)}_{[1,1]}, \left\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\right\}\right),
\end{aligned}$$

where (a) holds with equality when $\{\hat{\mathbf{D}}^{(t-1)}_{[k,\ell]}\}$ are optimum, (b) follows from the fact that each of $\{\hat{\mathbf{D}}^{(t)}_{[k,\ell]}\}$ is the optimal solution of the minimization in (38), (c) holds with equality when $\hat{\mathbf{T}}^{(t-1)}_{[1,1]}$ is optimum, and (d) is due to the fact that $\hat{\mathbf{T}}^{(t)}_{[1,1]}$ is the optimal solution of the minimization in (40).

2) Complexity Analysis: We derive the computational complexity of the proposed channel estimation method. Following the related work in [13], we measure the computational complexity by counting the number of complex multiplication (CM). Regarding the complexity of the estimation of column space in (27), we need to perform the eigenvalue decomposition, which requires $\mathcal{O}(M^3)$ computational complexity. Also, the dominant complexity of the proposed joint optimization in Algorithm 1 comes from the computations in (35), (39) and (41), which respectively require

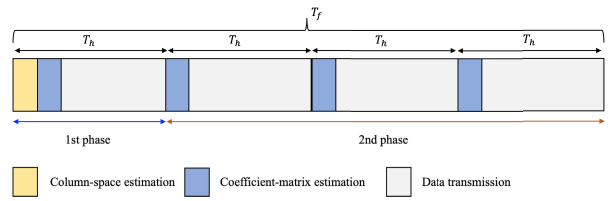

Fig. 4. The description of the proposed two-phase communication protocol.

$$\Delta_{\text{LS}} \triangleq N\left(\widehat{\text{rk}}(\mathbf{F})\right)^2 KL \tag{44}$$

$$\Delta_D \triangleq N(3\widehat{\text{rk}}(\mathbf{F}) + 1)KL \tag{45}$$

$$\Delta_T \triangleq N(\widehat{\text{rk}}(\mathbf{F}) + 2)KL. \tag{46}$$

Thus, the overall computational complexity of the proposed CLRA-JO is computed as

$$\Psi_{\text{ CLRA-JO}} = \mathcal{O}\left(M^3 + \Delta_{\text{LS}} + t_{\max}(\Delta_D + \Delta_T)\right). \tag{47}$$

Remark 4: The proposed channel estimation method can be efficiently operated under the two-phase communication protocol in Fig. 4. The key idea is based on the fact that the coherence time of the BS-RIS channel (denoted by T_f) is commonly much longer than that of the RIS-User channels (denoted by T_h). Accordingly, $\mathbf{S}_{\text{col}}$ and $\mathbf{T}_{[k,\ell]}$'s are changed every T_f and T_h time slots, respectively. Following this protocol, the overall training overhead becomes lower as T_f grows. Namely, in the second phase, we only require the NB_r training overhead, instead of $M_{\text{RF}}B_c + NB_r$. In our simulations in Section V, the effectiveness of the two-phase communication protocol will be demonstrated. Moreover, the two-phase communication protocol can reduce the computational complexity in (47) since in this case, the impact of the complexity $\mathcal{O}\left(M^3\right)$ can be negligible. ■

V. Simulation Results

In this section, we evaluate the performances of the proposed channel estimation method for RIS-assisted mmWave MU-MIMO systems with hybrid beamforming structures. In Section V-A, we will demonstrate the superiority of the proposed CLRA-JO for XL-RIS assisted XL-MIMO systems by comparing with the state-of-the-art (SOTA) methods in [13] and [31]. In Section V-B, we will verify the practicality of the proposed CLRA-JO via experiments on *real* 28GHz UPA channel data in [44]. Following the performance metric in the related works [12], [13], [28], [29], we adopt the normalized mean square error (NMSE) for the evaluation of channel estimation accuracy, defined as

$$\text{NMSE} \triangleq \mathbb{E}\left[\frac{1}{K}\sum_{k=1}^{K} \frac{\|\hat{\mathbf{H}}_{[\text{eff},k]} - \mathbf{H}_{[\text{eff},k]}\|_F^2}{\|\mathbf{H}_{[\text{eff},k]}\|_F^2}\right]. \tag{48}$$

The expectation is evaluated by Monte Carlo simulations with 10^3 trials.

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A. XL-RIS Assisted XL-MIMO Systems

We verify the superiority of the proposed channel estimation method (named CLRA-JO) for XL-RIS assisted multi-user XL-MIMO systems. To accomplish this, we evaluate its estimation accuracy in the three categories of wireless channels, defined in Section II: 1) *Far-far* field channels; 2) *Far-near* field channels; 3) *Near-near* field channels. In our simulations, we consider the MU-MIMO system with $M = N = 128 \times 2 = 256$, $L = 2 \times 2 = 4$, $N_{\mathrm{RF}} = 8$, the number of spatial paths $N_f = N_{h_k} = 3$, $\forall k \in [K]$, the system frequency $f = 50$ GHz and the wavelength $\lambda = c/f = 0.006\mathrm{m}$. Given the system wavelength λ, the RD and MIMO-RD are respectively determined as

$$Z_{\mathrm{RD}} = 98.33\mathrm{m} \text{ and } Z_{\mathrm{MRD}} = 196.66\mathrm{m}.$$

To build the far- and near-field channels, the distances between the BS and RIS are chosen as $z_f = 250\mathrm{m}$ and $z_f = 150\mathrm{m}$, respectively. Also, to create the far- and near-field channels, the distances between the RIS and the K users, denoted by $\{z_{h_k} : k \in [K]\}$, are uniformly distributed over $[120\mathrm{m}, 130\mathrm{m}]$ and $[20\mathrm{m}, 30\mathrm{m}]$, respectively. The signal-to-noise ratio (SNR) is defined as

$$\mathsf{SNR} = 10 \log \left(\frac{P}{\sigma^2} \right) [\mathrm{dB}], \tag{49}$$

where P and σ^2 are defined in (14) and (15), respectively. Without loss of generality, we set $\sigma^2 = 1$ and control P to reflect a channel quality. Regarding the modeling of wireless channels, it is assumed that the power is equally divided into each signal path as in [13]. Accordingly, the channel gains $\alpha_{h_k}^i$ in (9) and (10), and α_f^i in (12) and (13) are respectively defined as

$$\alpha_{h_k}^i = \sqrt{\frac{NL}{N_{h_k}}} e^{j\psi_{h_k}^i} \text{ and } \alpha_f^i = \sqrt{\frac{MN}{N_f}} e^{j\psi_f^i}, \tag{50}$$

where $\psi_{h_k}^i$ and ψ_f^i are chosen uniformly and randomly from $[0, 2\pi)$. In addition, the distance-dependent large-scale path loss is considered in Fig. 8. The angle of all AoDs and AoAs are chosen uniformly and randomly from $[0, 2\pi)$.

For comparison, the following benchmark methods are considered:

- **CLRA-LS**: In the proposed CLRA-JO, the coefficient matrices can be simply derived from the standard LS estimation (i.e., $\hat{\mathbf{T}}_k^{\mathrm{LS}}$ in (35)) instead of the joint optimization in Algorithm. This simplified method is called CLRA-LS, which is used to figure out the performance gain of the proposed joint optimization.
- **S-MMV**: The channel estimation problem is formulated as a multiple measurements vector (MMV) problem on the basis of the projected measurements on the common column space. The MMV problem is efficiently addressed via SOMP algorithm (see Algorithm 1 in [13]).
- **S-MJCE**: The effective channel is estimated via two-step (subspace) multi-user joint channel estimation procedure (see Algorithm 2 in [13]). The S-MJCE is considered as the best-known CS-based method.
- **3D-M-LAOMP**: The effective channel is estimated via the two-phase multi-user joint channel estimation procedure using 3D MMV-based look ahead OMP (3D-M-LAOMP) [31]. We note that this method is only applicable to the far-near field channels with ULA.

We point out that the S-MMV and S-MJCE are originally developed based on the far-far field channels with ULA. In order to secure more benchmark methods, they will be applied to the other categories of wireless channels with proper modifications (see Remark 5 and Remark 6 for details). Regarding the hyperparameters, the resolution of the quantization for dictionary matrices in the CS-based benchmark methods is set by $G = 512$, and in the proposed CLRA-JO, we choose the $B_r = 1$ and $t_{\max} = 10$ in Algorithm 1.

Remark 5: We emphasize that there is no prior work on the *near-near* field channels in the literature. In the existing *far-near field* channel estimation methods in [17], [20], [31], [32], and [45], it is assumed that the number of spatial paths in the BS-RIS and RIS-User channels (denoted by N_f and $\{N_{h_k} : k \in [K]\}$, respectively) are exactly known a priori. This unrealistic assumption makes them impractical. In addition, ULA was only considered as it is quite challenging to design a dictionary representing the *near-field* UPA response vectors or solve a 2D-ANM problem in a large-scale system. These methods cannot be used in our system model based on practical UPA. Also, their extension to UPA is non-trivial. To secure more benchmark methods, we extend the S-MJCE and 3D-M-LAOMP to the far-far and far-near field channels with UPA, respectively. The associated UPA dictionaries are constructed by means of the Kronecker product of the horizontal ULA dictionary and the vertical ULA dictionary, which is the common way to design a UPA dictionary [13], [45], [46]. Based on the S-MJCE and S-MMV, moreover, we develop the benchmark method for the near-near field channels with ULA, where the near-field dictionaries are designed following a similar way in [31]. They are further extended into the UPA systems using the aforementioned Kronecker product. ■

Remark 6: We remark that our near-field channels in Section II are modeled using the *practical* array response vector in (5), where an element-wise path loss (or channel gain) is taken into account. However, in the benchmark methods (i.e., the CS-based methods), the dictionaries are constructed using the *constant* array response vector [46], where a simplified constant path loss is assumed. Putting the constant path loss into (5), it is defined as follows:

$$\tilde{\mathbf{b}}_N(d, \theta, \phi) \triangleq \sqrt{\frac{1}{N}} \left[e^{-j\frac{2\pi}{\lambda}\Delta d(1,1)}, \ldots, e^{-j\frac{2\pi}{\lambda}\Delta d(N_h, N_v)} \right]^{\mathsf{T}}. \tag{51}$$

Due to this mismatch, the benchmark methods cannot perform well in our practical channel models. For comparison in their favorable channel models, in our simulations, we additionally consider the *simplified* near-field channels using the constant array response vector in (51). To clarify the channel models, in the below, the near-field channels using the constant and practical near-field array response vectors are called the *constant* and *practical* systems, respectively.

In addition, we conduct the simulations for the ULA-equipped MU-MIMO systems, where $M = N = 128$ and $L = 4$. This setting can ensure that RIS (or BS) apertures

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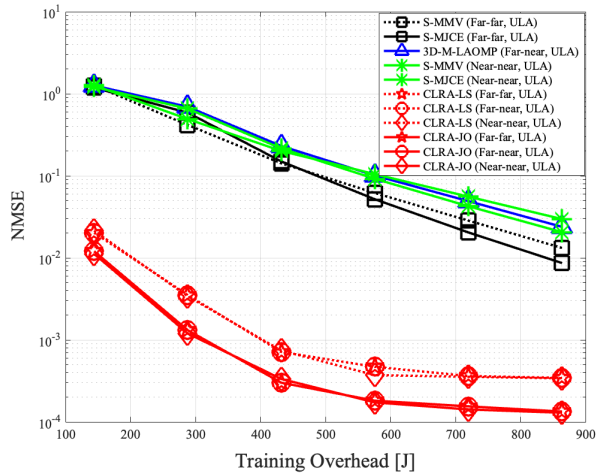

Fig. 5. The impact of the training overhead on the NMSE for ULA systems. $\mathsf{SNR} = 0$ dB and $K = 4$.

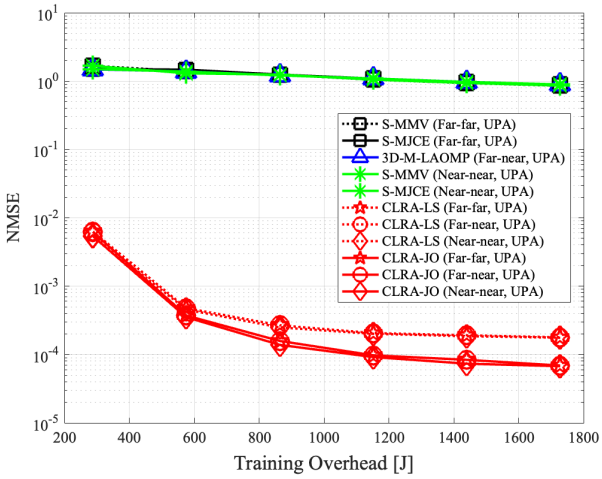

Fig. 6. The impact of the training overhead on the NMSE for UPA systems. $\mathsf{SNR} = 0$ dB and $K = 4$.

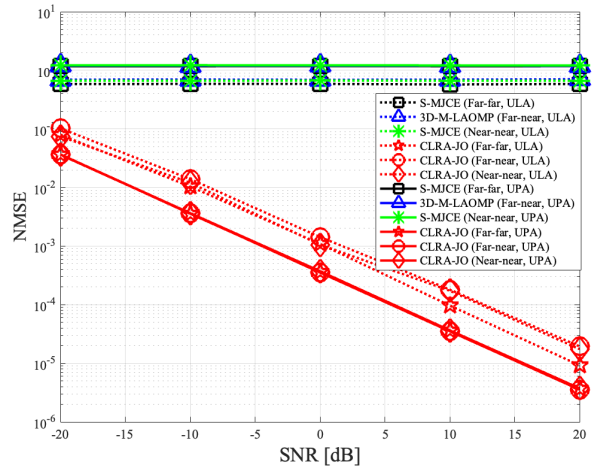

Fig. 7. The impact of SNR on the NMSE. $K = 4$ and $B_c = 10$.

for ULA and UPA are the almost same such as $(\lambda/2)128 \approx (\lambda/2)\sqrt{128^2 + 2^2}$. ■

Fig. 5 and Fig. 6 show the impact of training overhead on the NMSE for ULA- and UPA-equipped systems, respectively. In the both systems, the proposed CLRA-based methods can outperform the existing CS-based methods in all

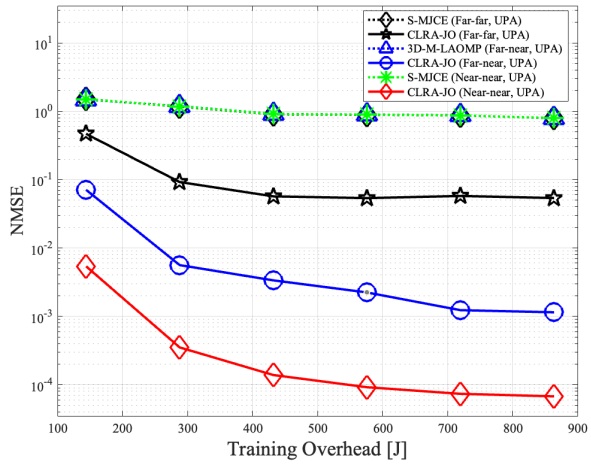

Fig. 8. The impact of the training overhead and distance on the NMSE. $K = 4$.

the categories of wireless channels. From the performance comparison between CLRA-JO and CLRA-LS, we can verify the effectiveness of the proposed joint optimization of user-specific coefficient matrices in Section IV-B. Noticeably, the proposed CLRA-JO can achieve higher channel estimation accuracy than the SOTA CS-based methods while reducing the pilot overhead about 80% (i.e., $J = 144$ and $J = 864$ for CLRA-based and CS-based methods, respectively). Moreover, the proposed CLRA-JO can show the almost similar performances regardless of the categories of wireless channels, which verifies that CLRA-JO is indeed the unified method. We point out that the CS-based methods are not practical as they perform worse in practical UPA than ULA due to the severe grid-mismatch problem.

Fig. 7 shows the impact of SNR on the NMSE. We first observe that the performance of CLRA-JO can improve as SNR grows. On the other hand, the performances of the CS-based methods are saturated due to the inevitable grid-mismatch problem. Note that such saturation level is determined according to the quantization resolution of steering vectors. Generally, it is impossible to reduce the saturation level at a desired accuracy (i.e., $\mathrm{NMSE} = 10^{-4}$) since the corresponding computational complexity is unmanageable. Together with the results in Fig. 6, it is verified that the proposed CLRA-JO is more suitable for practical UPA-equipped systems than the CS-based methods. From Fig. 6 an Fig. 7, we can derive the guideline to determine the hyperparameters B_c and B_r of CLRA-JO. Suppose that at operating $\mathsf{SNR} = 0\mathrm{dB}$, we aim to achieve an estimation accuracy $\mathrm{NMSE} = 10^{-4}$. From Fig. 6, it might be attained by only increasing the B_c and fixing $B_r = 1$. We can also see that using $B_c = 10$ (i.e., $J = 600$), the common column space is almost perfectly estimated since the performance improvement is not significant as the training overhead further increases. From the results in Fig. 7, we can figure out that SNR should be enhanced to attain the desired accuracy $\mathrm{NMSE} = 10^{-4}$. This gain can be easily obtained using the repeated symbol transmissions (i.e., the combining gain at the receiver). Toward this, the increment of the pilot overhead (i.e., from $B_r = 1$ to $B_r > 1$) should be accompanied.

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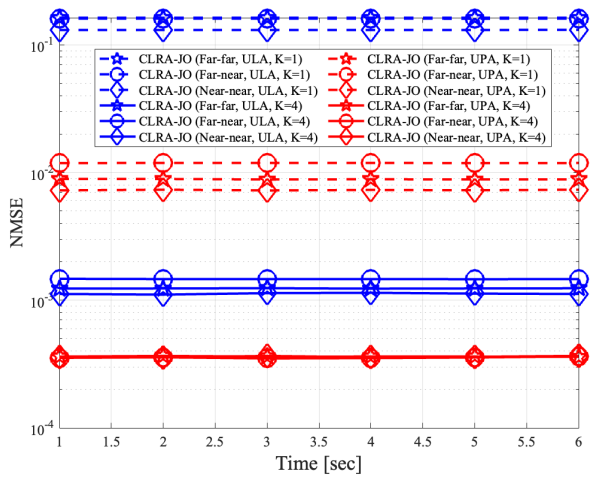

Fig. 9. The impact of the proposed two-phase communication protocol on the NMSE. $\mathsf{SNR} = 0$ dB and $B_c = 10$.

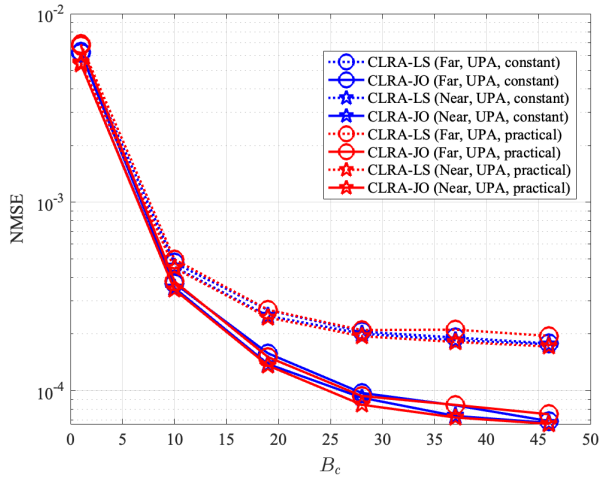

Fig. 10. The impact of the B_c on the NMSE for *practical* systems. $\mathsf{SNR} = 0$ dB and $K = 4$.

Fig. 8 shows the impact of training overhead and physical distances on the NMSE. Differently from Fig. 6, we consider the distance-dependent large-scale path loss $PL(z) = 10^{-2}z^{-2.2}$ with a distance z [47]. In Fig. 8, the channel gains of the far-far, far-near, and near-near field channels are determined according to the physical distances and $PL(z)$. As expected from the results in Fig. 7, the proposed CLRA-JO shows the best performance in the near-near field channel due to the shortest physical distance (i.e., the highest received SNR). Definitely, it is a natural consequence to be obtained if a channel estimation method is well-developed. In contrast, the CS-based methods cannot attain any performance improvement when the distance decreases. These results verify the superiority of the proposed CLRA-JO.

Fig. 9 shows the impact of the proposed two-phase communication protocol in Fig. 4 on the NMSE. In this simulation, the coherence times are set by $T_f = 6$ sec and $T_{h_k} = 1$ sec, $\forall k \in [K]$. As expected, the column space estimated only in the first phase (i.e., during the first 1 sec) can be effectively reused for channel estimations in the remaining second phase. Thus, as aforementioned, this protocol can reduce the computational complexity as well as the training overhead. Under the two-phase communication protocol, the proposed CLRA-JO requires about 2.8% computational complexity of S-MMV (i.e., the CS-based method having the lowest computational complexity). Here, the computational complexity of the S-MMV is provided in [13] such as

$$\Psi_{\text{S-MMV}} = \mathcal{O}\left(M^3 + N^2(\widehat{\text{rk}}(\mathbf{F})KL + G)\right).$$

We also note that the proposed CLRA-JO can indeed yield the multi-user gain since the performance with $K = 4$ outperforms that with $K = 1$. The performance gain is attained since the accuracy of the common column space estimation and the proposed joint optimization are improved as the K grows.

Fig. 10 depicts the estimation accuracy of the proposed methods in practical systems beyond the constant systems. In addition, we consider more realistic scenario where the users may experience different channel categories depending on their locations. For this, each RIS-User channel is determined uniformly and randomly from the far and near field channels. The BS-RIS channel is fixed as either far or near field channel (as indicated in Fig. 10). Since the benchmark methods cannot perform in these practical systems, we only evaluate the performances of the proposed CLRA-LS and CLRA-JO. We observe that the proposed methods can achieve almost the same performance regardless of the categories of wireless channels. This implies that the proposed methods can perform well as long as the low-rankness of the BS-RIS channel (i.e., the low-rankness of $\mathbf{F}$) is guaranteed. These results can support the practicality of the proposed methods.

Based on the above analysis, we can conclude that the proposed CLRA-JO would be a good practical candidate as the channel estimation method for XL-RIS assisted multi-user XL-MIMO systems.

B. RIS-Aided Massive MIMO Systems: 28GHz Channel Data

In this section, we demonstrate the effectiveness of the proposed CLRA-JO for the massive MIMO systems with $M = N = 8 \times 8 = 64$, $L = 2 \times 2 = 4$, and $N_{\text{RF}} = 8$. In the proposed method, we choose the $B_r = 1$ and $t_{\max} = 10$ in Algorithm 1. Also, the *real* 28GHz UPA channels are considered, in which the experimental data in [44] is used. Regarding the distances among the BS, RIS, and K users, $z_f = 50\text{m}$ and $\{z_{h_k} : k \in [K]\}$ are uniformly and randomly distributed over $[5\text{m}, 10\text{m}]$. Also, the distance-dependent path loss $PL(z)$ ($z = z_f$ or $z = z_{h_k}$) is modeled as

$$PL(z)\ [\text{dB}] = \alpha + 10\beta \log_{10}(z) + \epsilon, \tag{52}$$

where $\epsilon \sim \mathcal{N}(0, \nu^2)$. The channel parameters in (52) are set by $\alpha = 61.4$, $\beta = 2$, and $\nu = 5.8$ dB for the line-of-sight (LOS) paths, and $\alpha = 72.0$, $\beta = 2.92$, and $\nu = 8.7$ dB for the non-line-of-sight (NLOS) paths. From these parameters, the complex channel gains of $\{\mathbf{H}_k : k \in [K]\}$ and $\mathbf{F}$ are independently distributed according to $\mathcal{CN}(0, \gamma_{h_k}^2 10^{-0.1PL(z_{h_k})})$ and $\mathcal{CN}(0, \gamma_f^2 10^{-0.1PL(z_f)})$, respectively, where the normalization factor is determined as $\gamma_{h_k} = \sqrt{NL/N_{h_k}}$ and $\gamma_f = \sqrt{MN/N_f}$. In this real experimental data, the SNR defined

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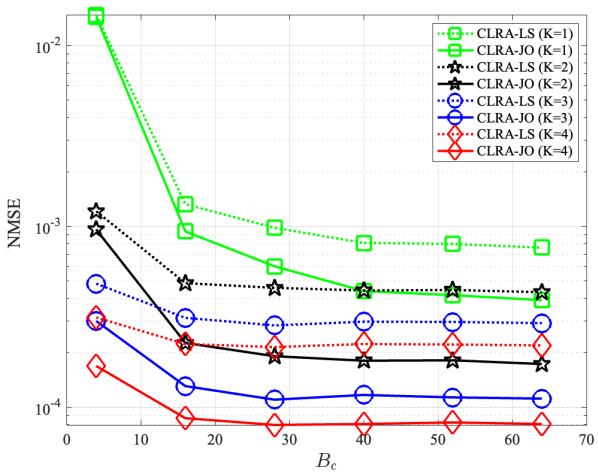

Fig. 11. The impact of B_c and K on the NMSE. $\mathsf{SNR} = 0$ dB, $N_f = 3$ and $N_{h_k} = 3$, $\forall k \in [K]$.

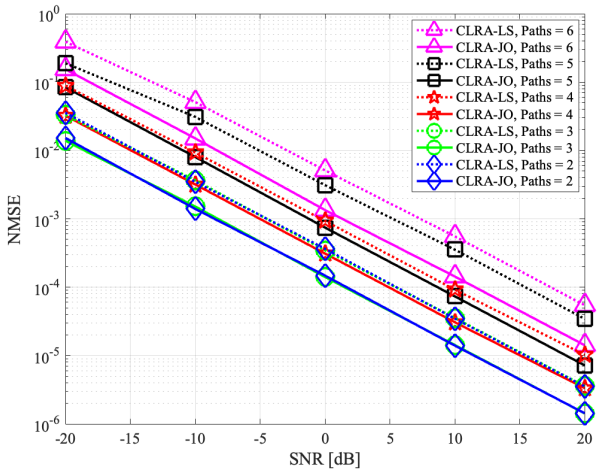

Fig. 12. The impact of SNR and N_f on the NMSE: $K = 4$, $B_c = 10$ and $N_{h_k} = 4$ $\forall k \in [K]$.

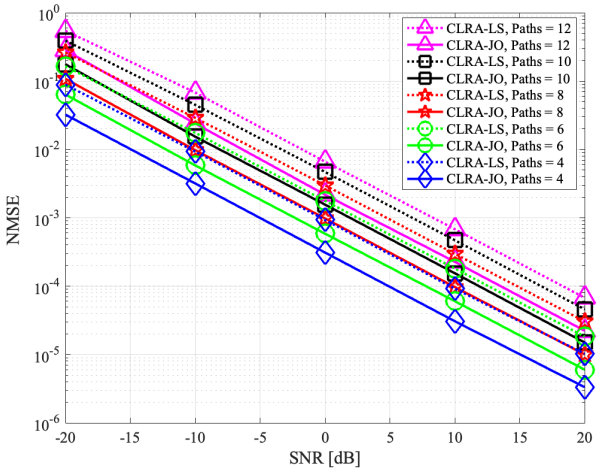

Fig. 13. The impact of SNR and N_{h_k}, denoted by Paths in label, on the NMSE: $K = 4$, $B_c = 10$ and $N_f = 4$.

in (49) is defined with

$$\sigma^2 = 10^{-0.1(PL(z_{h_k}+PL(z_f))}. \tag{53}$$

Fig. 11 shows the impact of training overhead B_c and the number of users K on the NMSE. From the comparisons with CLRA-LS, we can verify that our joint optimization in (37) indeed enhances the estimation accuracy of the user-specific coefficient matrices. We can see that the accuracy of the estimated $\mathbf{S}_{\text{col}}$ gets better as the number of users K grows. Remarkably, when the number of users is large (i.e., $K \geq 3$), the proposed channel estimation method can achieve an attractive performance with a very small training overhead. Thus, in practical MU-MIMO systems, the proposed method is very promising by considerably reducing the pilot overhead.

Fig. 12 and Fig. 13 show the impact of SNR and the number of signal paths on the NMSE. As expected, the estimation accuracy is enhanced as SNR increases (i.e., the noise perturbation in (35) becomes less). From the comparison with CLRA-LS, we can verify the effectiveness of the proposed joint optimization in Section IV-B, by alleviating the effect of lower SNR. Specifically, our joint optimization can attain the about $\mathsf{SNR} = 5$ dB gain compared with CLRA-LS. As expected, the estimation accuracy improves as N_f and N_{h_k} decrease. Thus, the proposed method is well-suited to the next-generation mmWave or THz communication systems, where the corresponding wireless channels consist of lower scattering and a smaller number of signal paths.

VI. Conclusion

We studied the channel estimation problem for XL-RIS assisted multi-user XL-MIMO systems with hybrid beamforming structures. In this system, we proposed the unified channel estimation method (named CLRA-JO) which can perform in the far- and near-field channels without any modification. Whereas, in the existing CS-based methods, dictionary should be designed by taking into account the characteristics of the near- and far-field channels. Via simulations and complexity analysis, it is demonstrated that the proposed CLRA-JO can yield better estimation accuracy than the state-of-the-art CS-based methods while having lower training overhead (e.g., the 80% reduction of the pilot overhead). Our on-going work is to design channel state information (CSI) feedback suitable for the proposed CLRA JO so that it can be applicable in frequency-division-duplexing (FDD)-based XL-RIS assisted multi-user XL-MIMO systems.

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